Experiment 2: Offline YOLO Detector Performance

Raw write-up for Chapter 4, Section 4.2

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Reporting cutoff: 7 November 2026

Purpose of this document. This is the raw, complete write-up of Experiment 2, written before the thesis chapter; Section 4.2 of the final report is extracted from it. Every number traces to a file under `output/` named in the "Reporting rules" paragraph of its slot.

4.2.1 Simulation

Status: completed. Four realistic-warehouse recordings, 12 698 left-camera frames, scored on 7 November 2026 against projected simulator ground truth with the four-class checkpoint (`weight/runs_1/detect/train/weights/best.pt`: bin, box, bucket, forklift), whose fourth class fires on the pallet jacks that are the moving objects of these recordings. Pooled precision 0.747; recall 0.502 over the truth boxes the exemptions keep and 0.288 over every truth box; moving-object recall 0.288. The walking people are simulator actors with no pose and no class, so they are outside the score on both sides.

Objective. Measure how well the project detector localises and names warehouse objects, independently of either SLAM integration, so that the dynamic filters of Experiment 3 have a known input quality; the moving-object recall is the headline, because it is the ceiling of any filtering benefit.

Process. `script/yolo_eval.py` projected the collision mesh of every model it has a class for into `cam0` from the simulator's pose stream to form truth boxes (the pallet jacks included, moving or parked), ran the checkpoint at confidence 0.35 with NMS at IoU 0.5 on the GPU, matched predictions greedily by class at IoU 0.5, and wrote per-frame and per-detection records; twelve overlays per recording were inspected before the aggregates were accepted. Truth boxes under 12 pixels or more than 70 % covered by nearer geometry are exempt (scored as neither hit nor miss), and predictions on unlabelled box-like scenery (loaded shelves, desks) are ignored. The three-class checkpoint of the retired evaluation is not used: it has no class for the moving objects. The earlier evaluation on the scripted worlds was retired on 6 November with its recordings and is not cited.

Datasets. The four `dataset_allsensor_realistic` recordings: `000` and `001` static (U and I layout), `002` and `003` with four pallet jacks on logistics jobs and four walking people. Ten pallet jacks stand in each world, so the forklift class has truth in every recording and *moving* truth only in `002` and `003` (3 333 and 4 562 moving truth boxes). Sources: `output/yolo_eval/dataset_allsensor_realistic_*_4class/` (`yolo_eval.csv`, `yolo_detections.csv`, `yolo_confusion.csv`, `overlay/`), `output/yolo_eval/summary_metrics.csv`, `output/compare/simulation/exp2-insights/`.

Reporting rules. Precision is over all retained predictions that were not ignored; recall is quoted on two bases, over the truth boxes the exemptions keep (the detector's task) and over every projected truth box (what is physically in view); moving-object recall counts a truth box as moving when its model moved more than 0.05 m between frames. Class metrics are conditional on a correct localisation.

Results. Table 2.1. The detector names what it finds and finds about half of what the exemptions leave it: pooled precision 0.747, recall 0.502 on the scored basis and 0.288 on the all-box basis, class F1 0.994 conditional on a match. Moving pallet jacks are found in 0.283 and 0.291 of their frames on `002` and `003`. The forklift class is the weakest on the all-box basis (recall 0.12 to 0.18 against 0.18 to 0.45 for the others, table 2.6), and the box class carries 34–70 % of the false positives.

Table 2.1: Detector scores per recording and pooled, four-class checkpoint, confidence 0.35, matching IoU 0.5. "Truth" counts every projected box and "exempt" those under 12 pixels or more than 70 % covered; R_{scored} is recall over the rest, R_{all} over every box, R_{mov} over the moving truth boxes.

Recording	Frames	Truth	Exempt	Pred.	TP	FP	P	R_{scored}	R_{all} / R_{mov}
`realistic_000`	3 805	38 401	16 157	16 673	10 940	2 693	0.802	0.492	0.285 / —
`realistic_001`	2 937	33 260	14 286	14 359	8 834	3 907	0.693	0.466	0.266 / —
`realistic_002`	2 409	28 049	10 907	13 002	8 821	2 332	0.791	0.515	0.314 / 0.283
`realistic_003`	3 547	40 498	18 235	19 148	11 852	4 757	0.714	0.532	0.293 / 0.291
Pooled	12 698	140 208	59 585	63 182	40 447	13 689	0.747	0.502	0.288 / 0.288

Analysis. The misses are a size effect and the false positives are near misses. Frame-level recall rises monotonically with the truth-box side, from 0.02 at 24–32 pixels to 0.64 above 192 pixels (table 2.2, figure 2.1); at object level every object is found in most of the frames where its box is large, and the moving pallet jacks follow the same curve as the parked ones, so the moving-object recall of 0.29 is what the pallet jacks' typical distance from the camera gives, not a motion effect. Of the unmatched predictions, 74–79 % overlap the right object below the IoU threshold and only 8–9 % land on nothing (table 2.3, figure 2.2): a looser matching threshold would move most false positives into the hits, and a filter that masks by box does not care which. The misses come as flicker rather than blindness: 39–47 % of the gaps between consecutive detections of one object are a single frame, and gaps longer than 2 s hold 20–32 % of the misses (table 2.4). Raising the confidence threshold buys precision at a steep price in recall: at 0.75 precision is 0.84 to 0.91 and moving-object recall 0.16 to 0.20 (table 2.5).

Limitations, each stated once. *Projected truth:* boxes are the projection of collision meshes, tighter than a human would draw and exempted by rules a human would not apply; the scored basis is the easier one and both are given. *People:* the walking people are actors without pose or mesh and outside the checkpoint's classes; no truth exists for them, their detections (if any) are charged as false positives, and a filter built on this detector cannot mask them at all. *One operating point:* a single confidence threshold was retained, so no average precision. *Checkpoint provenance:* the four-class checkpoint was trained on 8 October 2026 and its training set is not registered in this write-up.

Summary. The detector is reliable at naming and unreliable at finding: class F1 0.994 given a match, recall 0.50 over the boxes it is asked to find and 0.29 over everything in view, moving-object recall 0.29. The miss is a size effect and the false positives are mostly boxes a little off the right object. For the filters of Experiment 3 the input is a mask that covers under a third of the moving pallet jacks' frames and none of the people.

Future plan. Add a person class and give the actors projectable geometry, or annotate a frame subset by hand, so that the moving agents the filter exists for are all scorable; raise the input resolution or the

Table 2.2: Recall against truth-box size, pooled over the four recordings. Frame level: the frame's truth boxes are binned by the frame's mean truth-box side (square root of the mean truth area; the per-box truth is not logged), recall is TP/(TP+FN) of those frames. Object level: each object's median matched-box side; a miss is a frame inside a gap shorter than 2 s between two matches of that object, so objects that leave the view are not charged. Moving: object moving in at least half of its matched frames. Source: `recall_by_size.csv`.

Side [px]	frame level			object level			
	boxes	recall	R_{mov} / R_{stat}	objects mov/stat	frames	R_{mov}	R_{stat}
0–24	0	—	— / —	0/1	21	—	0.24
24–32	745	0.02	— / 0.02	1/1	932	0.96	0.93
32–48	30027	0.10	0.16 / 0.10	0/18	7120	—	0.79
48–64	29454	0.18	0.11 / 0.18	3/17	7453	0.89	0.72
64–96	37685	0.30	0.35 / 0.30	0/42	14312	—	0.80
96–128	21342	0.43	0.56 / 0.42	2/39	9856	0.53	0.85
128–192	17840	0.54	0.70 / 0.53	3/45	9524	0.87	0.85
192–∞	3115	0.64	0.60 / 0.64	0/14	2794	—	0.83

Table 2.3: What the unmatched predictions are, per recording. An unmatched prediction is a *duplicate* when it overlaps a matched prediction of the same class by IoU ≥ 0.5; a *near miss* when its best IoU with an unmatched truth box of its class is in $[0.2, 0.5)$; *slight overlap* when that IoU is in $(0, 0.2)$; *no overlap* when it touches no truth box. The scorer excludes the predictions that land on unlabelled scenery (*ignored*); they are counted here from the per-frame totals because the per-box file does not flag them, so the four categories sum to unmatched, not to scored false positives. "At border": the box touches an image edge. Source: `output/compare/simulation/exp2-insights/fp_anatomy.csv`.

Recording	unmatched	scored FP	ignored (scenery)	duplicate	near miss [0.2,0.5)	slight (0,0.2)	none 0	at border
`dataset_allsensor_realistic_000_4class`	5210	2693	3040	2 (0 %)	3994 (77 %)	745 (14 %)	469 (9 %)	12 %
`dataset_allsensor_realistic_001_4class`	5229	3907	1618	0 (0 %)	4330 (83 %)	626 (12 %)	273 (5 %)	13 %
`dataset_allsensor_realistic_002_4class`	3860	2332	1849	1 (0 %)	2860 (74 %)	684 (18 %)	315 (8 %)	13 %
`dataset_allsensor_realistic_003_4class`	6849	4757	2539	2 (0 %)	5101 (74 %)	1213 (18 %)	533 (8 %)	12 %

Table 2.4: Structure of the misses, per recording, from the gaps between consecutive matches of the same object (gaps under 2 s only). "Single-frame" is the share of gaps one frame long; "in long gaps" the share of the gap frames that lie in gaps of 1 s or more. "Short-gap misses / FN" compares the misses this reading finds with the scorer's total false negatives: the remainder are misses at the ends of tracks or in gaps longer than 2 s, which this reading cannot separate from objects leaving the view. "Objects under 0.5": objects with at least 30 frames whose in-track recall is below one half. Source: `miss_structure.csv`.

Recording	objects	gaps	median [frames]	p90 [frames]	single-frame	in long gaps	short-gap misses / FN
`dataset_allsensor_realistic_000_4class`	45 (4 under 0.5)	513	2	9	47 %	20 %	2038 / 27461 (7 %)
`dataset_allsensor_realistic_001_4class`	51 (12 under 0.5)	513	2	17	40 %	24 %	2952 / 24426 (12 %)
`dataset_allsensor_realistic_002_4class`	45 (7 under 0.5)	288	2	8	44 %	30 %	1330 / 19228 (7 %)
`dataset_allsensor_realistic_003_4class`	45 (4 under 0.5)	646	2	15	39 %	32 %	3658 / 28646 (13 %)

Table 2.5: Score-threshold sweep over the retained predictions. Precision, recall and moving-object recall if every prediction below the threshold were discarded; the logged operating point is 0.35, and nothing below it was retained, so the sweep can only go up. Matching is not re-run, so a discarded higher-scored box cannot free its truth for a lower-scored one; the values are therefore lower bounds on recall by a small margin. Source: `threshold_sweep.csv`.

Threshold	realistic_000			realistic_001			realistic_002			realistic_003		
	P	R	R_{mov}	P	R	R_{mov}	P	R	R_{mov}	P	R	R_{mov}
0.35	0.80	0.28	0.00	0.69	0.27	0.00	0.79	0.31	0.28	0.71	0.29	0.29
0.45	0.82	0.28	0.00	0.72	0.26	0.00	0.82	0.31	0.27	0.74	0.28	0.28
0.55	0.85	0.27	0.00	0.74	0.25	0.00	0.85	0.30	0.26	0.76	0.27	0.26
0.65	0.87	0.25	0.00	0.77	0.23	0.00	0.88	0.28	0.23	0.79	0.24	0.23
0.75	0.89	0.21	0.00	0.81	0.19	0.00	0.91	0.23	0.20	0.84	0.20	0.16
0.85	0.94	0.12	0.00	0.86	0.08	0.00	0.94	0.11	0.09	0.93	0.09	0.06

Table 2.6: Misses and false positives by class, per recording, from the scorer's class-agnostic confusion counts. Recall is the share of truth boxes of the class associated with any prediction; FP share is the class's share of the recording's background-row predictions. The pallet jack is not a class of the loaded 3-class weights and is an ignore region, so it does not appear. Source: `output/yolo_eval/dataset_allsensor_realistic_*_4class/yolo_confusion.csv`.

	realistic_000			realistic_001			realistic_002			realistic_003		
Class	truth	recall	FP share	truth	recall	FP share	truth	recall	FP share	truth	recall	FP share
bin	4219	0.30	0.09	1072	0.18	0.00	3038	0.40	0.05	1624	0.00	0.04
box	15541	0.22	0.34	19658	0.32	0.70	11161	0.45	0.46	20068	0.35	0.51
bucket	11428	0.44	0.29	6529	0.21	0.07	6515	0.26	0.19	10816	0.35	0.16
forklift	7213	0.18	0.27	6001	0.17	0.22	7335	0.12	0.30	7990	0.15	0.29

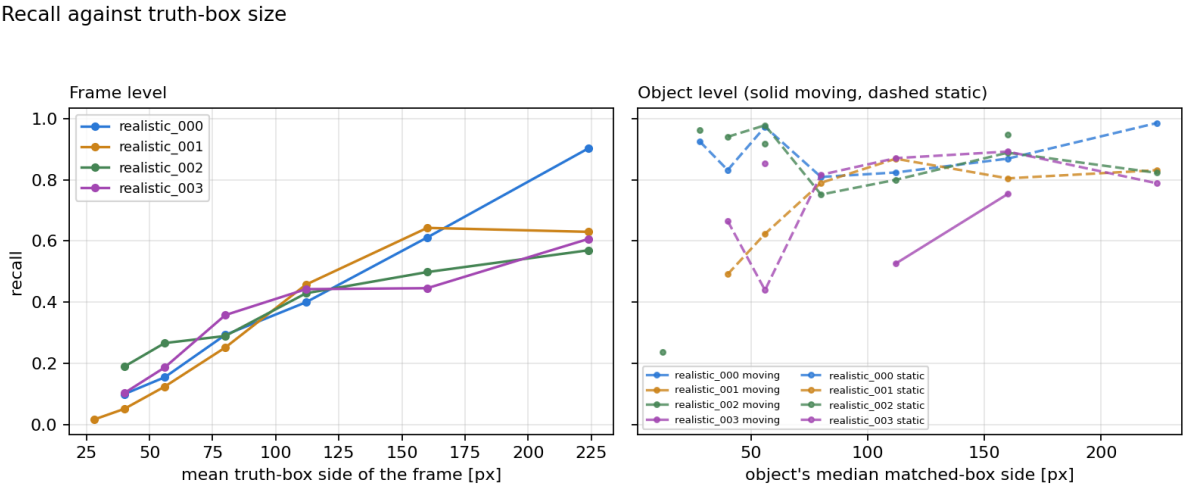


Figure 2.1: Recall against truth-box size: per frame (left, the frame's mean truth-box side) and per object (right, the object's median matched-box side; solid moving, dashed parked). The curves of the four recordings coincide, and the moving pallet jacks lie on the parked objects' curve.

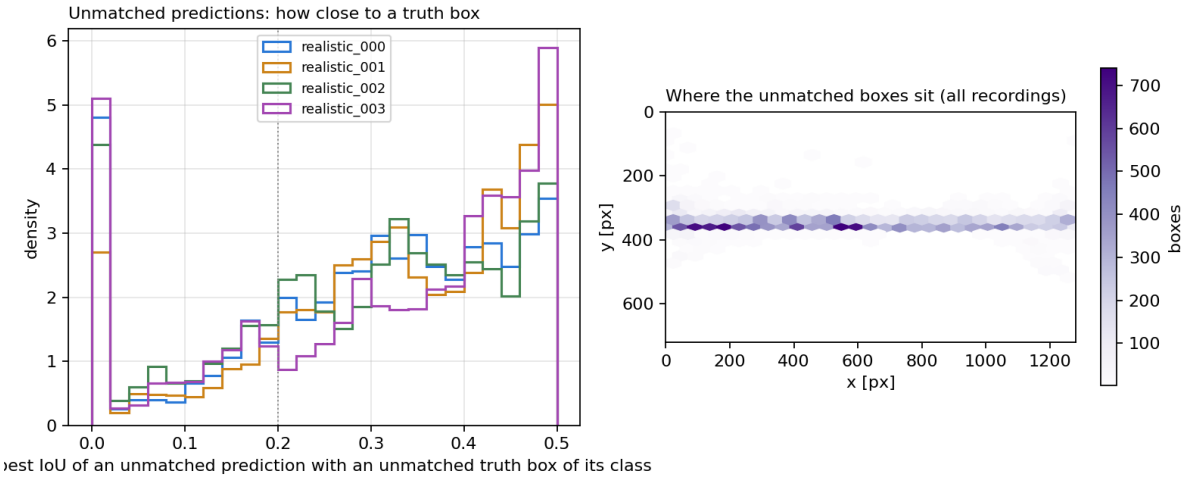


Figure 2.2: What the unmatched predictions are, per recording: near misses (right object, IoU 0.2–0.5), slight overlaps, boxes on nothing, and predictions ignored on unlabelled scenery.

training set's share of small boxes, since the miss is a size effect; and report a precision–recall curve from a run with the threshold at zero.

4.2.2 Real robot

Status: planned. Real imagery exists; no real ground-truth boxes do.

Reason. Projected ground truth needs object poses and meshes, which the real site does not provide. The four real recordings in use (`realv6_000`, `realv7_000–002`) hold the imagery, so the detector can be run on them, but nothing can score it until frames are annotated by hand.

Next action. Annotate a stratified frame subset of `dataset_realv6_000` and `dataset_realv7_000` to `_002` across the illumination levels of the test site, run the detector at confidence 0.35 and matching IoU 0.5, and report precision, recall, F1 and moving-object recall; the same annotated subset is the only route to a scored simulation slot as well, since the realistic worlds give the detector nothing projectable to find.